

$$2(a) \quad (f \circ g)(x) = 1/\sqrt{x+9}, \quad \text{Dom} = (-9, \infty)$$

$$2(b) \quad (g \circ f)(x) = 1/\sqrt{x} + 9, \quad \text{Dom} = (0, \infty)$$

$$2(c) \quad (f \circ f)(x) = x^{1/4}, \quad \text{Dom} = (0, \infty)$$

$$2(d) \quad (g \circ g)(x) = x + 18, \quad \text{Dom} = (-\infty, \infty)$$

$$3 \quad (f \circ g \circ h)(x) = 8\sin(x^2) - 9$$

$$4 \quad f(x) = x/(9+x), \quad g(x) = x^{1/3}$$

$$5(a) \quad g(g(g(4))) = 3$$

$$5(b) \quad (f \circ f \circ f)(5) = 6$$

$$5(c) \quad (f \circ f \circ g)(5) = 4$$

$$5(d) \quad (g \circ f \circ g)(1) = 1$$